

ECON 0150 | Economic Data Analysis

The economist's data analysis pipeline.

Part 5.3 | Numerical Control Variables

Pittsburgh Housing

What is the value of an additional square foot of living area?

House prices depend on:

- *Square foot*
- *Age of the house*
- *Number of bedrooms*
- *Neighborhood*
- *Many other factors!*

Pittsburgh Housing

What is the value of an additional square foot of living area?

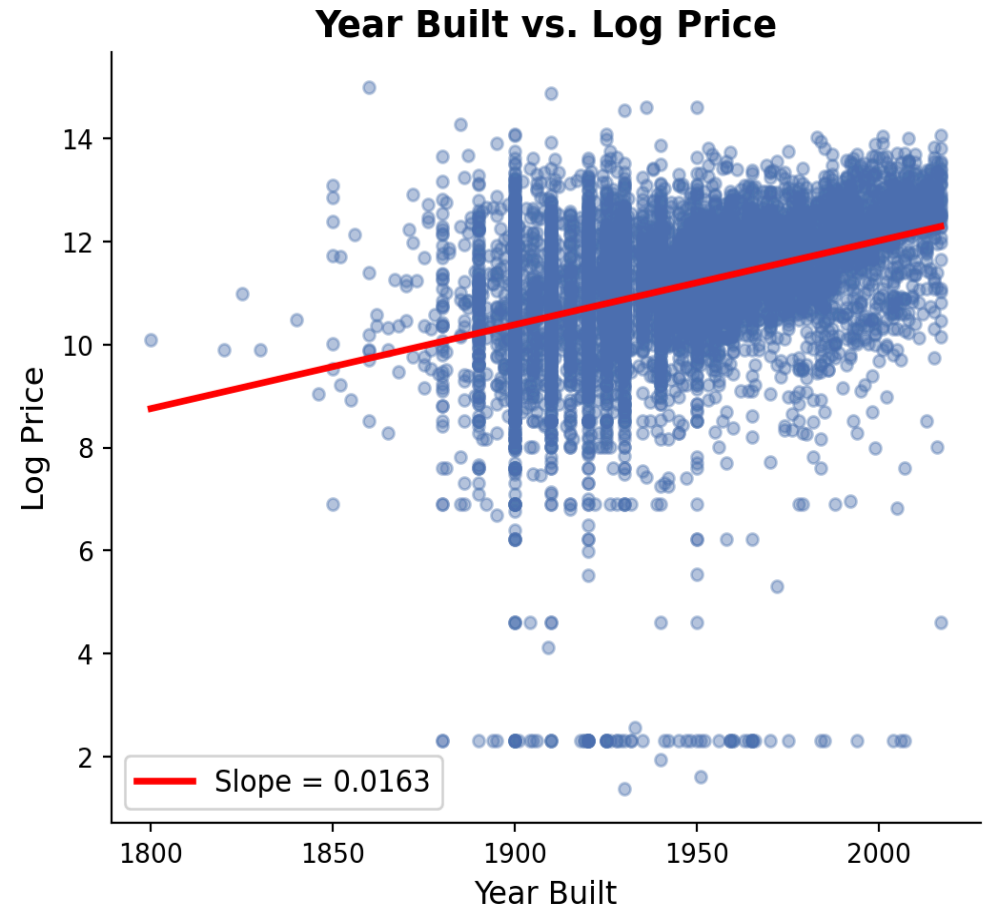
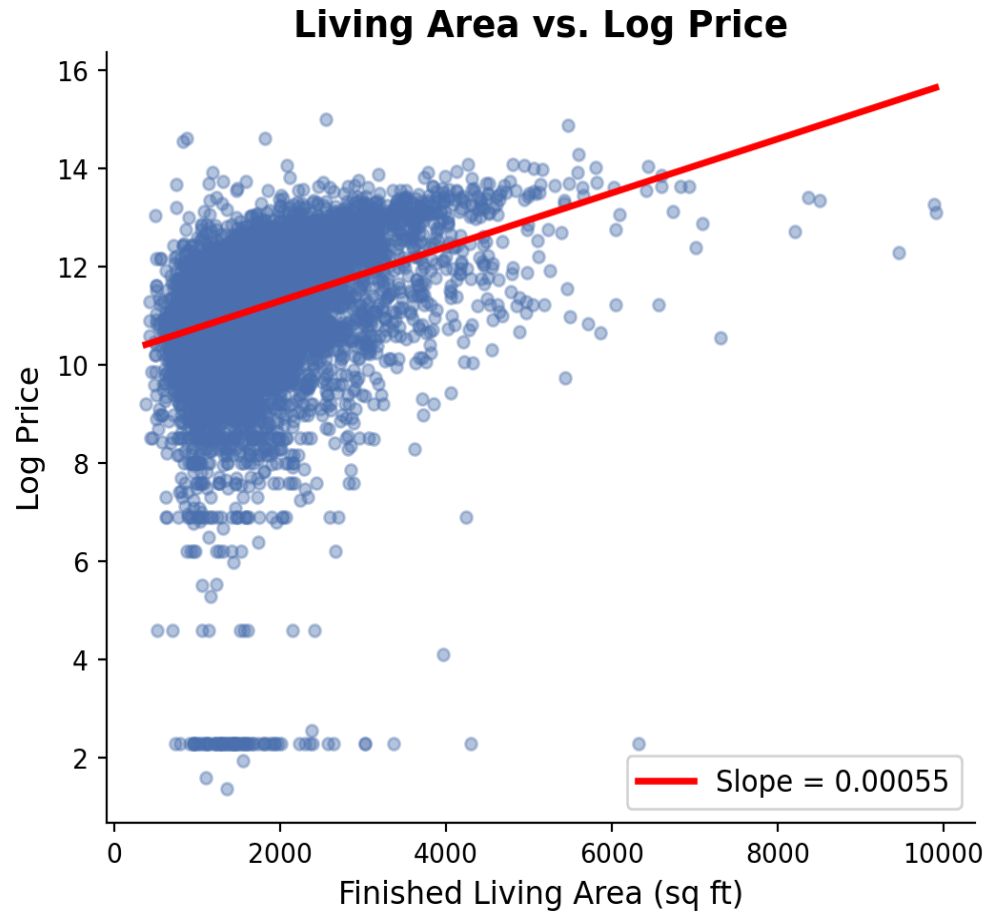
Pittsburgh housing data with 10,000 homes.

Variables:

- *SALEPRICE: Sale price of the home*
- *LOGPRICE: Log of sale price*
- *FINISHEDLIVINGAREA: Square footage of finished living space*
- *YEARBLT: Year the home was built*

Pittsburgh Housing: Separate Models

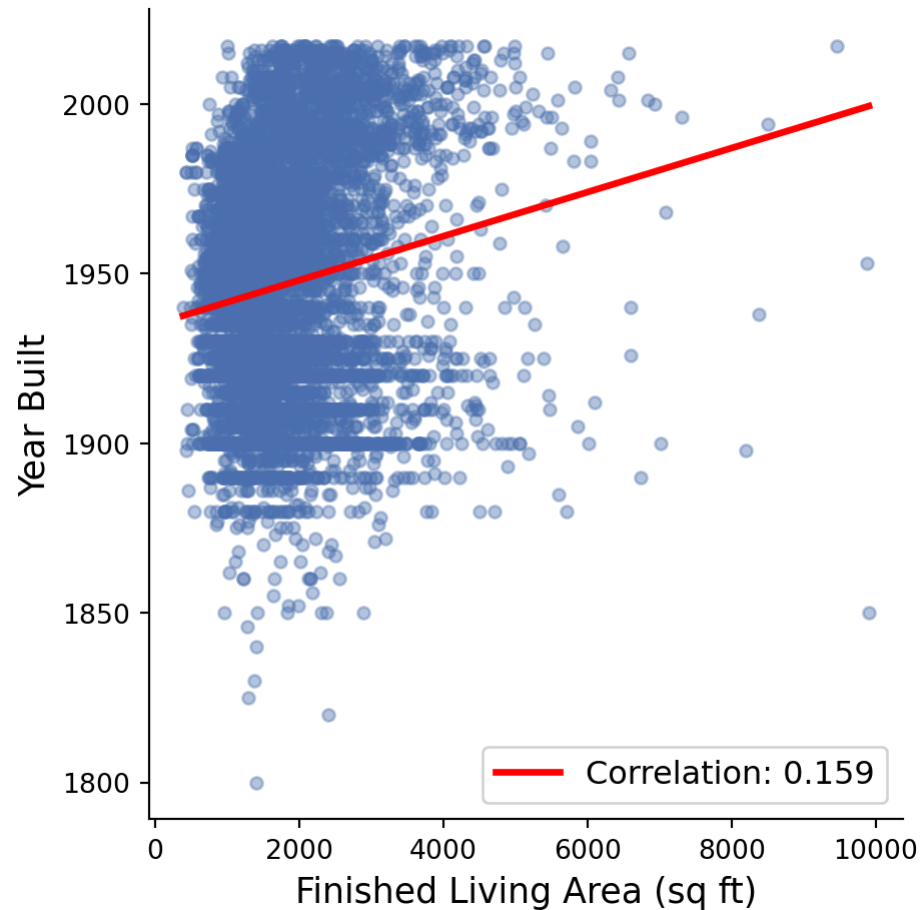
*Lets predict price using **Living Area** and **Year Built**.*



Both living area and year built have positive relationships with price.

Pittsburgh Housing: Related Predictor Variables

*Predictor variables **Living Area** and **Year Built** are themselves correlated.*

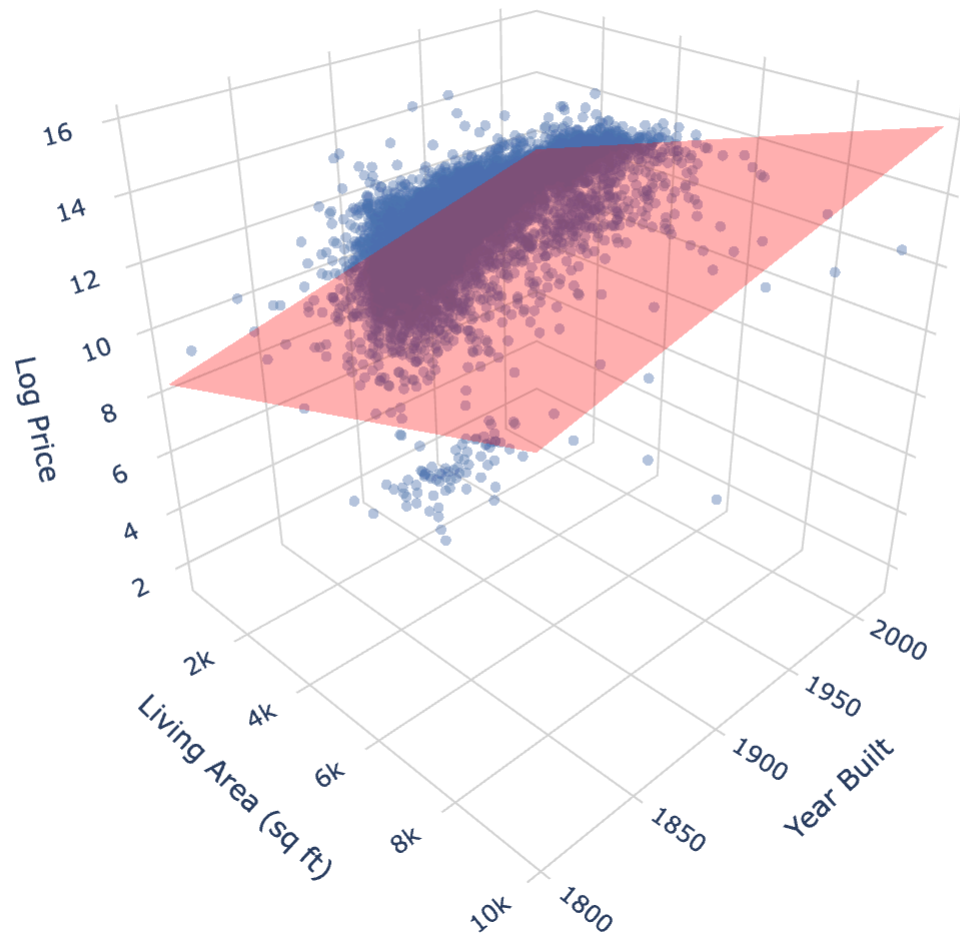


Larger homes tend to be built more recently.

So maybe larger homes are more valuable simply because they're newer....

Pittsburgh Housing: Multiple Regression

We can model multiple predictor variables simultaneously.



The Multiple Regression Equation

Extending the best-fitting line to multiple dimensions

Single Variable:

$$\text{LogPrice} = \beta_0 + \beta_1 \times \text{LivingArea} + \epsilon$$

Multiple Variables:

$$\text{LogPrice} = \beta_0 + \beta_1 \times \text{LivingArea} + \beta_2 \times \text{YearBuilt} + \epsilon$$

Interpretation:

- β_0 = *Base log price (intercept)*
- β_1 = *Effect of one more square foot, holding year built constant*
- β_2 = *Effect of being built one year later, holding living area constant*

Pittsburgh Housing

What is the value of an additional square foot?

Without controlling for year built:

- *We compare small (old) houses to large (new) houses*
- *The size effect includes the age effect*
- *Overestimates the true value of square footage*

With controlling for year built:

- *We compare houses of similar age but different sizes*
- *Isolates the pure size effect*
- *Gives us the true value of square footage*

*This is what we call a **control variable**.*

Multiple Regression with Control

We can adjust for multiple variables simultaneously.

Model 1: Living Area Only

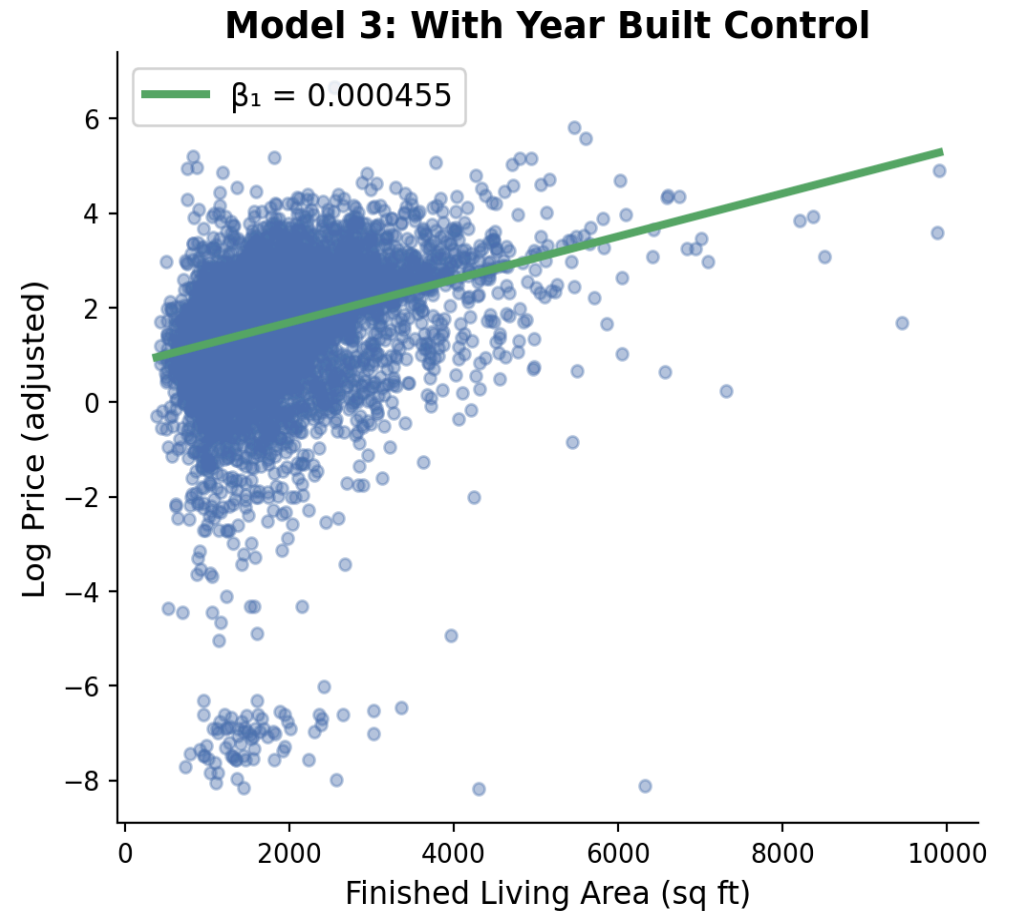
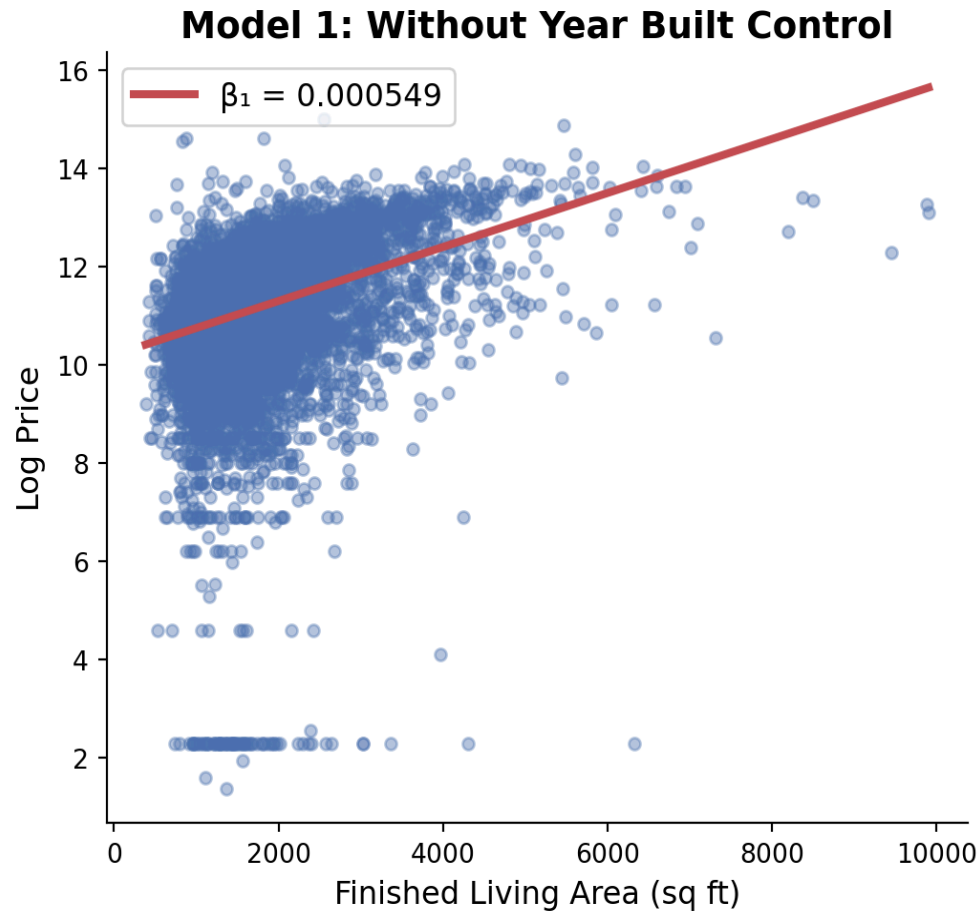
	coef	std err	t	P> t	[0.025	0.975]
Intercept	10.2029	0.031	328.476	0.000	10.142	10.264
FINISHEDLIVINGAREA	0.0005	1.65e-05	33.260	0.000	0.001	0.001

Model 3: Living Area + Year Built Control

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-17.9337	0.741	-24.213	0.000	-19.386	-16.482
FINISHEDLIVINGAREA	0.0005	1.56e-05	29.095	0.000	0.000	0.000
YEARBLT	0.0145	0.000	38.018	0.000	0.014	0.015

Comparing Models

Compare coefficients with and without the control.

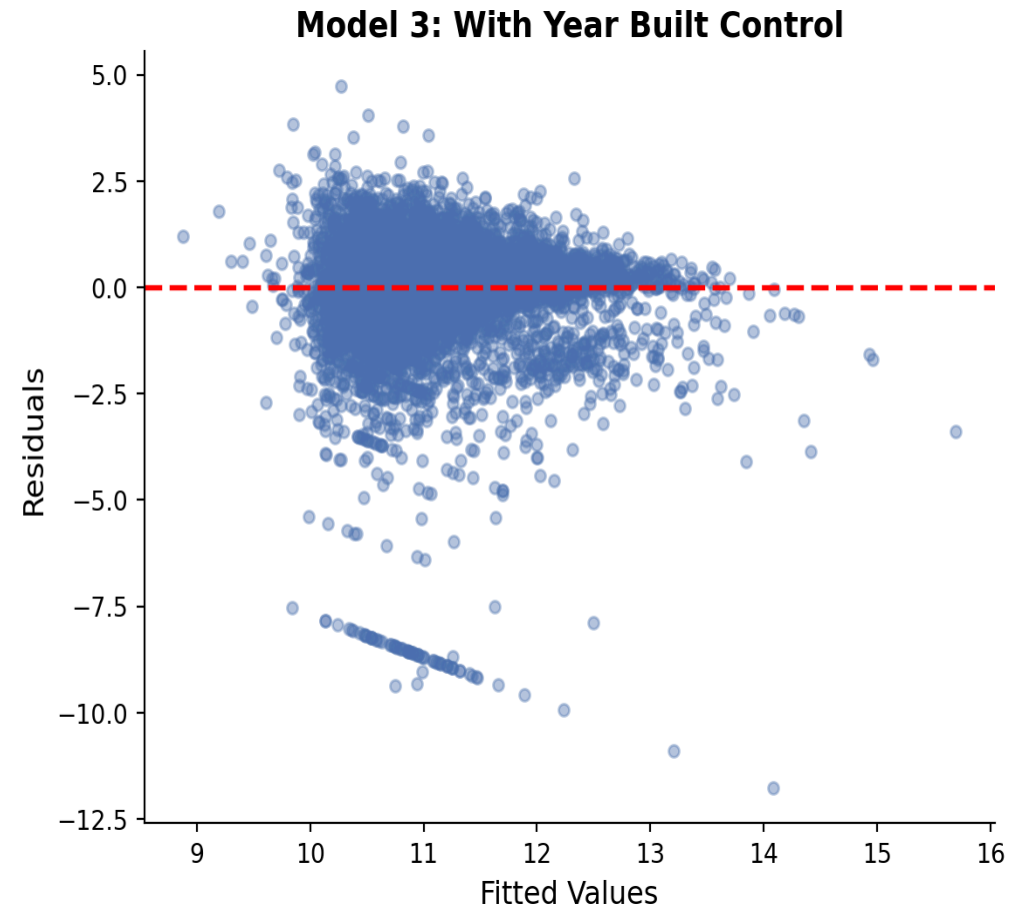
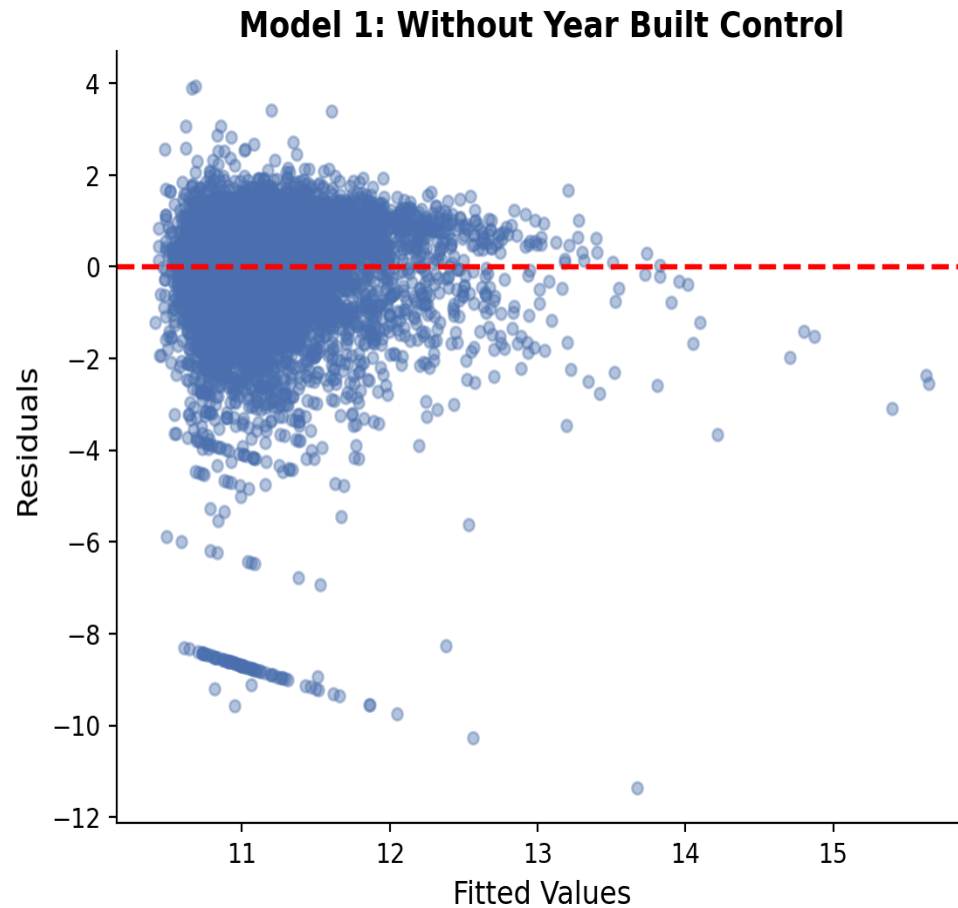


Without control: size effect was inflated by age effect

With control: we get the size effect between houses of equal age

Residual Diagnostics

Checking if our model improved



Residuals look somewhat more random with the control.

Log Interpretation

For log outcomes, coefficients represent proportional changes.

For small coefficients (< 0.1):

$$\beta \times 100 \approx \text{percentage change}$$

Interpretation (holding year built constant):

- Each additional sq ft increases price by 0.045%
- 100 additional sq ft increases price by 4.55%

Interpretation (holding living area constant):

- Each year newer increases price by 1.454%
- A house 10 years newer increases price by 14.54%

The “Holding Constant” Intuition

A control variable compares two similar observations.

Without control (simple regression):

$$\text{LogPrice} = \beta_0 + \beta_1 \times \text{LivingArea} + \epsilon$$

- *Compares all small houses to all large houses*
- *Large houses are newer, so β_1 captures both size AND age effects*

With control (multiple regression):

$$\text{LogPrice} = \beta_0 + \beta_1 \times \text{LivingArea} + \beta_2 \times \text{YearBuilt} + \epsilon$$

- *Compares small and large houses built in the same year*
- *β_1 now captures only the pure size effect*
- *β_2 captures only the pure age effect*

Summary

Main ideas about numerical control variables

1. ***Control variables*** help isolate the effect of your main predictor
2. ***“Holding constant”*** means comparing observations with similar control values
3. ***Coefficients change*** when you add controls because you remove confounding
4. ***Choose controls carefully***: confounders yes, mediators no
5. ***Check residuals*** to see if controls improved the model

Exercise 5.3 | Pittsburgh Housing Prices

Lets find the value of an extra square feet in the Pittsburgh housing market.

Step 1: Load and Explore the Data

```
1 # Load the data
2 data = pd.read_csv(file_path + 'pgh_housing.csv', index_col=0)
3 data['LOGPRICE'] = np.log(data['SALEPRICE'])
4 data.head()
```


Step 2: Modeling Relationships Separately

```
1 # Model 1: Living area only
2 model_area = smf.ols('LOGPRICE ~ FINISHEDLIVINGAREA', data=data).fit()
3 print(model_area.summary().tables[1])
```

```
1 # Model 2: Year built only
2 model_year = smf.ols('LOGPRICE ~ YEARBLT', data=data).fit()
3 print(model_year.summary().tables[1])
```

Step 3: Compare Living Area and Year Built

```
1 # Plot the relationship between living area and year built
2 plt.figure(figsize=(8, 5))
3 sns.scatterplot(data, y='FINISHEDLIVINGAREA', x='YEARBLT', alpha=0.5)
4 plt.title('Relationship Between Living Area and Year Built')
5 plt.ylabel('Finished Living Area (sq ft)')
6 plt.xlabel('Year Built')
```

Step 4: Multiple Regression with Control Variable

```
1 # Model 3: Both variables (with control)
2 model_both = smf.ols('LOGPRICE ~ FINISHEDLIVINGAREA + YEARBLT', data=data).fit()
3 print(model_both.summary().tables[1])
```

Step 5: Checking Residuals

```
1 # Create residual plots for both models
2 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
3
4 # Model 1: Without control
5 residuals_area = model_area.resid
6 predictions_area = model_area.fittedvalues
7 ax1.scatter(predictions_area, residuals_area, alpha=0.5, color='#4C72B0')
8 ax1.axhline(y=0, color='red', linestyle='--', linewidth=2)
9 ax1.set_title('Model 1: Without Year Built Control')
10
11 # Model 3: With control
12 residuals_both = model_both.resid
13 predictions_both = model_both.fittedvalues
14 ax2.scatter(predictions_both, residuals_both, alpha=0.5, color='#4C72B0')
15 ax2.axhline(y=0, color='red', linestyle='--', linewidth=2)
16 ax2.set_title('Model 3: With Year Built Control')
```